

Capital Asset Pricing and Arbitrage Pricing Theory

Chapter 4 - Theoretical Overview of Asset Pricing Models

[Theoretical Overview: Capital Asset Pricing and Arbitrage Pricing Theory | Springer Nature Link](#)

Objectives

- Understand the Capital Asset Pricing (CAP) Model
- Learn the key components and assumptions of CAPM
- Explore the Fama-French multi-factor models
- Understand Arbitrage Pricing Theory (APT)
- Compare CAPM and APT models
- Recognize applications in portfolio management

Asset Pricing Models

Purpose:

- Value financial assets
- Estimate cost of capital
- Make investment decisions
- Understand risk-return trade-off

Foundational Models:

- **Capital Asset Pricing Model (CAPM)**
 - Single-factor model
 - Introduced by Sharpe (1964) and Lintner (1975)
- **Arbitrage Pricing Theory (APT)**
 - Multi-factor model
 - Developed by Stephen Ross (1970s)

History of CAPM

Year	Event
1964	Sharpe introduces CAPM
1964-1975	Lintner contributes to CAPM development
Late 1960s-70s	Practical implementation begins
1976	Vanguard 500 Index Fund introduced
1970s-80s	Empirical testing of CAPM
1980s	CAPM widely taught and adopted

Significance:

- First systematic model of risk/return trade-off
- Foundation for modern portfolio theory
- Basis for Fama-French multi-factor models

CAPM Overview

Definition: Model that quantifies the trade-off between risk (beta) and expected return

Key Insight:

- Investors are compensated for systematic risk only
- Idiosyncratic risk can be diversified away
- Relationship between risk and return is linear

Core Equation:

$$ER_i = R_f + \beta_i(ER_m - R_f)$$

Where:

- ER_i = Expected return of asset
- R_f = Risk-free rate
- β_i = Systematic risk relative to market
- ER_m = Expected market return

Beta as a Risk Measure

What is Beta?

- Measures stock's sensitivity to market movements
- Amplifies or dampens market returns

Mathematical Formula:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m}$$

- $\sigma_{i,m}$ = Covariance between stock and market
- σ_m^2 = Variance of market returns
- $\rho_{i,m}$ = Correlation between stock and market

Beta	Interpretation
$\beta = 1$	Moves with the market
$\beta > 1$	More volatile than market (aggressive)
$\beta < 1$	Less volatile than market (defensive)
$\beta = 0$	No correlation with market

CAPM Components

Risk-Free Rate (R_f)

- Government bonds (e.g., US Treasury bills)
- Represents time value of money
- Minimum return for any investment

Expected Market Return (ER_m)

- Broad market index (e.g., S&P 500)
- Represents average market performance

Market Risk Premium ($ER_m - R_f$)

- Additional return above risk-free rate
- Compensates for taking market risk

Beta (β)

- Measures systematic risk
- Specific to each asset

Expected Return Example

Example:

- Investment has 50% chance of gaining 20%
- Investment has 50% chance of losing 10%

Calculation:

- $ER = (0.5 \times 20\%) + (0.5 \times -10\%)$
- $ER = 10\% - 5\% = 5\%$

Interpretation:

- Expected return = 5%
- Weighted average of all possible outcomes
- Not guaranteed, just expected

Excess Return Formulation

Original Equation:

$$R_i = R_f + \beta_i(R_m - R_f)$$

Excess Return Form:

$$R_i - R_f = \alpha + \beta_i(R_m - R_f)$$

Interpretation:

$R_i - R_f$ = Excess return of stock

$R_m - R_f$ = Excess return of market

α = Intercept (Jensen's alpha)

β_i = Slope (risk measure)

What Alpha Measures:

Positive alpha = Outperformance

Negative alpha =

Underperformance

Alpha = 0 = Fairly priced

Why is CAPM Important?

1. Theoretical Framework

- Understand relationship between return and systematic risk
- Helps investors make informed decisions

2. Asset Valuation

- Compare expected return with required return
- Identify underpriced or overpriced securities

3. Cost of Equity Estimation

- Crucial for capital budgeting
- Influences investment decisions

4. Market Equilibrium

- Expected returns aligned with risk
- Explains market dynamics

5. Financial Metrics

- Foundation for alpha (excess returns)
- Foundation for beta (systematic risk)

CAPM Assumptions

Investor Assumptions:

- Investors are rational and risk-averse
- Investors hold diversified portfolios
- Investors can borrow and lend at risk-free rate

Market Assumptions:

- Markets are efficient
- All investors have same information
- No taxes or transaction costs
- No inflation or interest rate changes

Implications:

- Only systematic risk matters
- Idiosyncratic risk is eliminated
- All investors hold the market portfolio

Limitations of CAPM

1. Unrealistic Assumptions

- Markets are not perfectly efficient
- Taxes and transaction costs exist
- Investors are not always rational

2. Single Factor Limitation

- Only considers market risk
- Ignores other important factors

3. Beta Instability

- Beta changes over time
- Different estimation periods give different results

4. Empirical Failures

- Small stocks outperform
- Value stocks outperform growth stocks
- Momentum effect exists

5. Black Box Beta

- Does not explain WHY beta matters
- No theoretical basis for risk premium

Fama-French 3-Factor Model

Motivation: CAPM fails to explain certain market anomalies

3 Factors:

Market Factor ($R_m - R_f$)

- Same as CAPM
- Excess return of market

Size Factor (SMB)

- Small Minus Big
- Small caps outperform large caps

Value Factor (HML)

- High Minus Low
- Value stocks outperform growth stocks

Formula:

$$\begin{aligned} R_i - R_f \\ = \alpha_i + \beta_{im}(R_m - R_f) + \beta_{iSMB}SMB \\ + \beta_{iHML}HML + \epsilon_i \end{aligned}$$

Fama-French 4-Factor Model

4 Factors:

- Market Factor ($R_m - R_f$)
- Size Factor (SMB)
- Value Factor (HML)
- Momentum Factor (MOM)

Momentum Effect:

- Past winners continue to outperform
- Past losers continue to underperform
- Captured by Carhart (1997)

Formula:

$$\begin{aligned} R_i - R_f &= \alpha_i + \beta_{im}(R_m - R_f) \\ &+ \beta_{iSMB}SMB + \beta_{iHML}HML \\ &+ \beta_{iMOM}MOM + \epsilon_i \end{aligned}$$

Fama-French 5-Factor Model

5 Factors:

- Market Factor ($R_m - R_f$)
- Size Factor (SMB)
- Value Factor (HML)

Profitability Factor (RMW)

- Robust Minus Weak
- High profitability vs low profitability

Investment Factor (CMA)

- Conservative Minus Aggressive
- Conservative investment vs aggressive investment

Formula:

$$\begin{aligned} R_i - R_f &= \alpha_i + \beta_{im}(R_m - R_f) + \beta_{iSMB}SMB \\ &\quad + \beta_{iHML}HML + \beta_{iRMW}RMW \\ &\quad + \beta_{iCMA}CMA + \epsilon_i \end{aligned}$$

Fama-French Models Comparison

Factor	3-Factor	4-Factor	5-Factor
Market ($R_m - R_f$)	✓	✓	✓
Size (SMB)	✓	✓	✓
Value (HML)	✓	✓	✓
Momentum (MOM)		✓	
Profitability (RMW)			✓
Investment (CMA)			✓

Arbitrage Pricing Theory (APT)

Definition: Multi-factor model that considers multiple sources of risk

Developed by: Stephen Ross (1970s)

Key Insight:

- Asset returns influenced by multiple factors
- Not just market risk
- Arbitrage opportunities drive prices

Core Principle:

$$E(R_i) = R_f + \beta_{i1}F_1 + \beta_{i2}F_2 + \dots + \beta_{ij}F_j$$

Where:

- $E(R_i)$ = Expected return of asset i
- R_f = Risk-free rate
- β_{ij} = Sensitivity of asset i to factor j
- F_j = Risk premium of factor j

Common APT Factors

Macroeconomic Factors:

- Inflation Rate
- Interest Rates
- GDP Growth
- Industrial Production
- Consumer Confidence

Market Factors:

- Market Returns
- Oil Prices
- Exchange Rates
- Credit Spreads

Industry-Specific Factors:

- Sector Performance
- Commodity Prices
- Regulatory Changes

Key Features of APT

1. Multi-Factor Approach

- Recognizes multiple risk sources
- More comprehensive than CAPM

2. Arbitrage-Based

- Exploited by arbitrageurs
- Drives prices to equilibrium

3. Flexible Framework

- No fixed factors specified
- Investors can choose relevant factors

4. Empirical Foundation

- Based on observed market relationships
- Data-driven factor selection

5. No CAPM Assumptions

- No need for market portfolio
- No need for risk-free borrowing/lending

Assumptions of APT

1. Efficient Markets: Prices reflect all available information

2. No Arbitrage

- Arbitrage opportunities quickly exploited
- Equilibrium prices established

3. Linear Relationship

- Returns described by linear function
- Multiple factors combined linearly

4. Asset-Specific Risk

- Idiosyncratic risk exists
- Can be diversified away

5. Factor Identification

- Factors are exogenous
- Factors affect all assets

Advantages of APT

1. More Realistic

- Multiple risk factors recognized
- Better captures real-world dynamics

2. Flexible

- Can choose relevant factors
- Adaptable to different contexts

3. No Market Portfolio

- No need for proxy market portfolio
- Avoids market portfolio issues

4. Empirical Foundation

- Based on observed relationships
- Data-driven factor selection

5. Captures Anomalies

- Explains CAPM anomalies
- Value, size, momentum effects

APT Disadvantages and Challenges

1. Factor Selection

- Determining relevant factors is challenging
- Requires extensive data analysis

2. Complexity

- More complex than CAPM
- Harder to implement practically

3. Data Requirements

- Needs more data for factor estimation
- Longer estimation periods needed

4. No Factor Guidance

- Doesn't specify which factors to use
- Can lead to data mining

5. Estimation Issues

- Factor sensitivities may be unstable
- Different factors for different periods

Comparison: CAPM vs APT

Feature	CAPM	APT
Risk Factors	Single (market)	Multiple (macroeconomic)
Sensitivity (Beta)	One beta	Multiple betas
Formula	$ER_i = R_f + \beta_i(ER_m - R_f)$	$E(R_i) = R_f + \sum \beta_{ij} F_j$
Ease of Use	Simpler	More complex
Real-World Use	Widely used	Advanced applications
Theoretical Basis	Portfolio theory	No arbitrage
Market Portfolio	Required	Not required
Factor Identification	Fixed	Flexible

Applications of CAPM and APT

CAPM Applications:

- **Portfolio Management**
 - Constructing efficient portfolios
 - Performance evaluation
- **Corporate Finance**
 - Cost of equity estimation
 - Capital budgeting
- **Asset Valuation**
 - Security pricing
 - Fair value assessment

APT Applications:

- **Advanced Portfolio Management**
 - Multi-factor strategies
 - Risk factor analysis
- **Hedge Fund Management**
 - Factor exposure analysis
 - Risk decomposition

Which Model to Use?

Choose CAPM When:

- Simple, single-factor analysis needed
- Market risk is the primary concern
- Limited data availability
- Regulatory requirements

Choose APT When:

- Multiple risk sources exist
- Complex portfolio strategies
- Advanced risk management needed
- Factor-based investing approach

Choose Fama-French When:

- Value or size effects matter
- Need better explanatory power
- Academic research context
- Cross-sectional analysis

Questions & Discussion

Review Questions:

- What does beta measure in CAPM?
- What are the key assumptions of CAPM?
- How does Fama-French 3-factor model differ from CAPM?
- What is the core principle of APT?
- When would you choose APT over CAPM?

Discussion Points:

- Limitations of single-factor models
- Practical challenges of multi-factor models
- Role of arbitrage in asset pricing
- Relevance of CAPM in modern finance